NeuroScan AI — Multimodal Stroke Screening Report

GMU MASE 2026 AI Healthcare Competition | Model Version: 1.0.0-MASE2026 | Generated: 2026-11-02 09:14:22 UTC

IMPORTANT CLINICAL & REGULATORY LIMITATION: NeuroScan AI is an experimental research and educational software prototype developed for the GMU MASE 2026 competition. It is NOT a certified or clinically validated medical device and must never be used to independently diagnose, treat, or rule out stroke or any neurological pathology. 1) Any sudden neurological symptom (Facial droop, Arm weakness, Speech abnormality) must trigger immediate emergency medical escalation regardless of AI output. 2) Non-contrast CT (NCCT) sensitivity for hyperacute ischaemic stroke is limited; a negative CT haemorrhage result NEVER rules out ischaemic stroke. 3) Normal vital signs (BP, HR, SpO2) never exclude acute cerebrovascular events.

Assessment ID:	ASM-DEMOC003	**Patient Ref ID:**	DEMO-SCENARIO-C
Patient Age:	74	**Symptom Onset:**	Unknown (Wake-up presentation)
Record Provenance:	SIMULATED_DEMO RECORD	**Assessment Timestamp:**	2026-11-02T09:14:22

1. Integrated Urgency Assessment & Clinical Recommendation

URGENCY CLASSIFICATION: **IMMEDIATE_EMERGENCY_ESCALATION**

RECOMMENDED NEXT ACTION: IMMEDIATE EMERGENCY ESCALATION: Activate emergency stroke protocol (Call Emergency Services / Stroke Team) immediately. Do not delay care for additional uploads. Note: Negative CT haemorrhage never excludes ischaemic stroke.

2. FAST Neurological Screening & Vital Signs

FAST Domain	Manual Observation	Experimental AI Status	Vital Sign	Recorded Value & Provenance
Facial Weakness (F)	UNSURE	POOR_LIGHTING_REJECTED (score: None)	Blood Pressure	Not Recorded [SIMULATED_DEMO]
Arm Weakness (A)	NO	NOT_PERFORMED	Heart Rate	76 bpm
Speech Difficulty (S)	YES	See Acoustic Module	SpO2 / Temp	SpO2: 97.0% | Temp: N/A C

3. AI Brain CT Scan Analysis (Research Decision Support — Not a Diagnosis)

No Brain CT scan was uploaded for this assessment. Missing imaging never delays symptom-based emergency escalation.

4. Multimodal Synthesis, Safety Escalations & Limitations

Reasons for Escalation:	Sudden speech impairment reported during FAST screening.
Cross-Modal Conflicts:	No cross-modal conflicts detected.
Missing Information:	Definitive Manual Facial Assessment, Valid Webcam Facial Frame (POOR_LIGHTING_REJECTED), AI Pose Arm Drift Screening, Acoustic Speech Recording Analysis, Blood Pressure (Systolic/Diastolic), Non-Contrast Brain CT Scan
Uncertainty & Limitations:	Webcam Facial AI unreliable/rejected (POOR_LIGHTING_REJECTED); relying on manual clinical check.; No Brain CT scan uploaded. Symptom triage proceeds immediately without waiting for imaging.